

Describing Welfare

Session 2

Ethan Ligon

September 15, 2026

Introduction

Deaton is in your reading/ folder on the hub, or PDF.

- Deaton, chs. 3 and 4
- Deaton & Zaidi, *Guidelines for Constructing Consumption Aggregates*
 - Deaton & Zaidi §§2–4 if you read nothing else
- **Christiaensen, Ligon, & Sohnesen (2022)**: at least memorize the title.

Measuring Individual Welfare

Many notions of individual welfare; generally lead to some unidimensional index, say w (bigger is better):

- Subjective ("How happy are you?")
- Utility (How is this different from Subjective happiness?)
- Material (Something about command over economic resources)
 - Wealth?
 - Income?
 - Consumption expenditures?

Aside: What about risk?

Material Well-Being: Statistics

Some summary statistic of the *distribution* F of w in the population.

Inequality E.g., Atkinson's (1971) utilitarian notion of inequality:

$$I = U \left(\int_{\underline{w}}^{\bar{w}} w dF(w) \right) - \int_{\underline{w}}^{\bar{w}} U(w) dF(w) \geq 0$$

Poverty A quantile of some function $\phi_z(w)$; e.g.,

$$P(z) = \int_{\underline{w}}^z \phi_z(w) dF(w)$$

These are distributions *of the population*, so in practice those sample weights will matter!

Conventional options for w

Income

- Difficult to measure if sources are agricultural or informal
- Also highly variable—a farmer's main income may arrive once a year, after harvest, and be very risky.

Wealth

- What is it?
- Perhaps something like discounted expected income?
- But if income is difficult to measure then all possible future incomes may be much more difficult

Conventional options for w , continued

Assets

- Related to wealth, but may be measurable now
- Financial assets + Bicycles + Goats + House = ?? In principle these could be priced, but this may not be at all practical.

Consumption Expenditures

The "Gold standard"

- Probably easier to measure than income
- Consumption is smoother than income
- And closer to the theoretical object (permanent income, lifetime resources) that we really want.

The Theoretical Object

The household's problem

What is the consumption aggregate a measure of? Assume:

- (A1) Preferences** $U(c)$ over J goods, increasing, strictly concave, the same every period, discounted at β .
- (A2) One decision maker** the household maximizes a single U .
- (A3) One asset** a risk-free bond, gross return R , no borrowing limit.
- (A4) Expectations** $\mathbb{E}_t$ over the true conditional distribution.

With A_t resources on hand and $x_t = p_t \cdot c_t$ total expenditure,

$$V_t(A_t, p_t) = \max_{c_t} U(c_t) + \beta \mathbb{E}_t V_{t+1}(A_{t+1}, p_{t+1}),$$

subject to $A_{t+1} = R(A_t - x_t) + y_{t+1}$.

The marginal utility of expenditure

Differentiate. With $u_j = \partial U / \partial c_j$, the first-order condition in c_j and the envelope condition in A are

$$u_j(c_t) = \lambda_t p_{tj}, \quad \lambda_t \equiv \beta R \mathbb{E}_t[\partial V_{t+1} / \partial A] = \partial V_t / \partial A.$$

So λ_t is two things at once, which is what makes it useful:

- the multiplier on the period budget constraint, the utility value of one more cedi spent *today*;
- the derivative of the *value function*, the utility value of one more cedi of resources, the whole future included.

V is concave in A , so λ falls as the household gets richer.

Write $w_t = -\log \lambda_t$, which therefore rises with welfare. This is another candidate for the w of the opening slide.

The martingale condition

Substituting the envelope condition into the definition of λ_t ,

$$\lambda_t = \beta R \mathbb{E}_t \lambda_{t+1}.$$

When $\beta R = 1$ the marginal utility of expenditure is a martingale:

$$\mathbb{E}_t \lambda_{t+1} = \lambda_t.$$

A statement about information: whatever the household knew at t is already in λ_t , so nothing dated t forecasts the change.

- Needs A1–A4 *and* an interior solution: no binding borrowing limit.
- With several assets, $\beta \mathbb{E}_t[(\lambda_{t+1}/\lambda_t) R_{t+1}^j] = 1$ for every asset j held in positive quantity.

Why total expenditure?

Three steps, each with its own assumption. Let

$v(x, p) = \max\{U(c) : p \cdot c \leq x\}$, so $\lambda = \partial v / \partial x$.

1. At *given* prices x orders welfare, since $\lambda > 0$. Nothing beyond A1 is needed, but this ranks within a market only.
2. To rank across markets we need a scalar $P(p)$ with $v(x, p) = \psi(x/P(p))$. Such a P exists if and only if U is homothetic (A5), and then $P = e(p)$, the cost of a unit of the aggregate. Quasi-homothetic (A5'): two indices, a line $a(p)$ and a deflator $b(p)$.
3. So $\lambda = \psi'(x/e(p))/e(p)$, decreasing in real expenditure.

Under A5, real total expenditure $x/e(p)$ is a sufficient statistic for within-period welfare. It is what the rest of this session builds.

Hall's random walk

The strict permanent income hypothesis adds:

(A8) Constant prices or a martingale in p independent of λ , which puts the walk in quantities.

(A9) Quadratic utility $U(x) = -\frac{1}{2}(\bar{x} - x)^2$, so $\lambda_t = \bar{x} - x_t$ is *linear* in expenditure.

(A10) $\beta R = 1$ with R constant and known.

The martingale in λ is then a martingale in x :

$$\mathbb{E}_t[\bar{x} - x_{t+1}] = \bar{x} - x_t \implies \mathbb{E}_t x_{t+1} = x_t.$$

Consumption is a random walk (Hall 1978): nothing dated t forecasts Δx_{t+1} , lagged income included. Testable, but it needs longitudinal data.

The assumptions, collected

	Assumption	What it buys
A1–A2	Time-separable U , one maximizer	a well-defined λ
A3	One asset, no borrowing limit	an interior solution
A4	Rational expectations	$\mathbb{E}_t$ is the truth
A5	Homotheticity, or quasi-	a deflator, or a line and one
A6	Iso-elastic cardinalization	w affine in log real x
A7	Durables weakly separable	nondurables and services
A8–A10	Constant p , quadratic U , $\beta R = 1$	Hall's random walk

Everything from here to the end of the session is an attempt to measure x . None of it repairs an assumption in this table.